Dhawal Khatri | CE23B063 | INDIAN INSTITUTE OF TECHNOLOGY MADRAS

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institution	%/CGPA	Completion
B.Tech in Civil Engineering	Indian Institute of Technology, Madras	7.91/10	2027
XII – CBSE	Cambridge Court High School, Jaipur	92%	2023
X – CBSE	Cambridge Court High School, Jaipur	93.1%	2021

• Achieved **Bronze Medal** in SASMO Olympiad Qualifier in Mathematics 2022 - 2023 while in **12th** grade	2023
• Achieved the Position of **Ramanujan of the year** for the outstanding performance in Mathematics in School	2023
• Achieved **99.55th percentile** in JEE Mains, reflecting consistent computational aptitude and logical reasoning.	2023
• **Inter IIT Tech 14.0**: Secured a **Bronze Podium** finish across IITs, engineering rapid high-throughput multi-modal model	2025
• **Jane Street Poker Bot:** Competed against a global participant pool, successfully securing 13 percentile tournament finish	2026
• **Inter IIT Civil Conclave**: Achieved **Gold Podium** finish against all IIT, demonstrating excellence **spatial risk modeling**	2026
• **Codeforces Rank:** Secured a peak rating of **1302**, demonstrating advanced algorithmic and rapid problem-solving skill.	2026
• **Research**: Authored a new earthquake hazard modeling framework, currently under peer review a top **Elsevier journal**.	2026
• Achieved **Top 6 rank out of 11,852** competing teams across all IITs in the AMEX Codestreet 2026 National Hackathon	2026

SKILLS

Languages: C,Cpp, Python, JS, TS, Dart, Java	**Databases:** DBMS, Mongo, Postgres	**Frontend:** ReactJS, NextJS, Flutter, HTML
Deployment: AWS, Vercel, Render	**Tools:** Git, GitHub, Postman, MS Office	**Designing:** Figma, CSS, Bootstrap, Tailwind

PROFESSIONAL EXPERIENCE

Internships	**Frontend Developer** E-Subhalekha (Dec'24-Jan'25)	• Built and maintained **5+ web modules** using modern, **enhancing UI responsiveness** and **user experience**. • Delivered **100% of project tasks on time**, contributing to the deployment of **3 major client-facing features**. • Demonstrated strong attention to detail by fixing **20+ UI bugs**, improving app stability, and performance. • Built **10+ responsive UI** layouts using **HTML, CSS, and vanilla JavaScript**, ensuring browser compatibility. • Converted **Figma mockups** into **pixel-perfect pages** with cross-browser support and responsive behavior. • Developed **interactive landing pages** using **PHP**, handling both **frontend styling and backend scripting**
	Flutter Developer Chimertech (Nov'24-jan'25)	• Supported development **of 3+ agri-healthcare prototypes,** accelerating their product readiness for **field trials**. • Designed **5 experiment batches** ofdevices, contributing to a **30% increase** in detection accuracy and consistency. • Designed **scalable product** modules in Flutter, **focusing on UI/UX**, smooth navigation, and performance. • Integrated **REST APIs** and implemented **efficient state management** using **industry-standard patterns**. • Handled **confidential client** applications with **strict adherence to data** and **code protection protocols**.
	Subject Matter Expert Kautilya Education (May'24-jn'24)	• Authored **100+ advanced physics solutions** with **<1% error rate**, ensuring quality content for NEET/JEE aspirants. • Increased team efficiency by **30%** through **structured project management** and pro-active issue resolution. • Managed a team of **6+ interns**, assigning tasks and deadlines, which led to **25% faster content delivery** pipeline. • Delivered 5 robust **full-stack web** modules for Kautilya Education using **modern web frameworks**, improving **UI response** and load performance while keeping every module **maintainable, reusable**, and **easy to extend.** • Synchronize **frontend/backend** with **shared data layer**, end stale-data i**nconsistency** during active sessions
	Forward Developer Intern Zetta Tech LLP *(May'26-July'26)*	• Architected the **React Native** field-sales app end-to-end (offline recording, live GPS, Salesforce-linked workflows), authore **90 commits** and **25 merged** PRs across **4 releases** to Play Store, TestFlight for **120+ agent** • Design production **AI pipeline** (FastAPI, Celery, LLM calling) converting 3,000+, **40-minute** field recordings into scored, **structured CRM records**; resolved **silent transcript truncation** on audio via **chunked transcription**. • Root-caused production outage in which the 30s periodic task deep-scanning a 771,000-row table blew its time limit and silently broke **CRM write-back** for weeks; **rectified** with an indexed window query and review policy.
Free Lancing	**TNMSS (Dec'24-April'25)**	• Achieved a **40%** improvement in user experience by designing a responsive, user-friendly interface using Flutter. • Enabled **1,000+ teachers** to flag student mental health issues via red alert system for early psychologist referral. • Built real-time dashboards to track and support quicker, data-driven decisions by educators and psychologists. • Developed a **cross-platform mental health app** for the **Tamil Nadu Government** using **Flutter (Dart/Java).** • Implemented **dynamic UI** components, **routing, and data binding** in Flutter for a seamless user experience. • Used **JavaScript** to **handle backend interactions and manage external APIs** effectively and efficiently. • Contributed to system integration with native modules, ensuring scalable and maintainable architecture**.**
	IITM Mobility (Dec'24-march'25)	• Tracked **10,000+ trips** from **500+ users**, identifying high-demand zones and peak travel hours across campus. • Enabled **15% optimization** in bus schedules by aligning routes with real-time user trip data and demand patterns. • Reduced average wait **times by 20%** by analyzing **huge trip data** and refining timings at **10+ high-traffic bus stops**. • Developed a cross-platform **mobility app** using **Flutter (Dart)** for seamless campus-wide tracking and interaction. • Integrated 24/7 location tracking to support infrastructure planning in collab with the civil engineering department. • Designed dynamic Flutter UI components for smooth navigation and live updates based on location and user interaction. • Connected with backend using JavaScript, ensuring secure and efficient communication for real-time mobility operations.

PROJECTS and ACHIEVEMENTS

Jane Street Poker bot	• Architected an **exploitative** heads-up **poker agent in Python** for an **auction-variant engine**, building a **five-stage decision pipeline** that chains **Monte Carlo equity, opponent profiling, range discounting** and an **EV gate** • Validated a **full strategy rewrite across 18 controlled** 1,000-hand matches, proving the **two variants** were statistically **indistinguishable at 95%** confidence and **overturning a first-match result** that **looked decisive**. • Benchmarked the decision path against a **hard 30 ms per hand time budget**, measuring a **27 ms worst case**, and **re-keyed an adaptive throttle** that had only **engaged once 83% of the entire match time** bank was already gone.
Inter IIT Tech Meet 14.0	• **Constructed** the **speech-to-intent tool-calling** layer routing spoken prompts into **8+ automated editor actions**, resolving **relative commands** such as **'make it a bit warmer'** against current state rather than **absolute values**. • **Shipped** content-provenance tagging **stamping device**, **timestamp** and **last edit into EXIF**, and hand-wrote a Dart to Kotlin platform channel exposing **Android FLAG_SECURE to block screenshots and screen recording** entirely. • **Designed** a radial **multi-light control deck laying items** out along a parabola, with a **direct-manipulation transform** gizmo, **normalised coordinates, swipe-to-delete** and **zero hardcoded pixel dimensions** anywhere
Inter IIT	• 3 geospatial APIs (**Earth Engine terrain data**, **SoilGrids soil composition**, **live rainfall data**) fused into one **Flutter+Flask** pipeline, computing **real-time Factor-of-Safety** scores across Uttarakhand's 14,780 landslide zones. • 14,780 **active landslide zones** and a **national 3rd-place** risk ranking drove the decision to scope **the MVP** to

Civil Conclave	Uttarakhand first, validating **tap-to-query** risk scoring on **1 high-risk state** ahead of a planned **19-state rollout**. • 4.2 lakh km^2 of **landslide-prone terrain** spread across **19 Indian states** and **union territories framed** a clear case for replacing slow, **batch-oriented** manual **GIS surveys** with an instant, **tap-based field-scoring tool overall**.
BHASA PSHA APP* *(paper under review)* *Dr. STG Raghukanth*	• **Co-authored** the full research paper documenting both the **earthquake hazard** framework and its mobile implementation, submitted to ***Advances in Engineering Software*** (Elsevier, IF ~4.0) and currently **under review**. • Created the full visualization and **export layer rendering hazard curve**, uniform **hazard spectra** and **DSHA** response spectra across **20** built-in GMPE, with CSV/ PNG export feeding direct into structural design workflow • Engineered the **Dart-to-Java** platform-channel bridge feeding into **the native Java hazard** engine using scalar-only payloads, background-thread dispatch, holding interface response through heavy **PSHA integration** loops
Agentic AI & Structural Analysis *Dr. STG Raghukanth*	• Developed an **Agentic AI system** use Large Language Models and **OpenSees** to automate complex structural model and evaluate building behavior with vary **seismic, wind, and self-loads** via **dynamic LLM tool calling** • Built a **dynamic LLM pipeline** converting user inputs of building heights and base into a full structural model. • Automated OpenSees structural analysis by mapping LLM tool calls to evaluate multiple load factors accurately.
American Express CodeStreet 2026	• Built a Temporal.io durable-execution saga for airline rebooking, hotel booking, and a ground transfer with pre-registered LIFO compensation and idempotent rollback, validated against induced non-retryable carrier failure • Authored and ran 250,000-case event-level Monte Carlo simulation in Python that directly isolate the search-breadth versus allocation lever behind same-day recovery, byte-reproducible end to end under fix random seed • Authored an OPA Rego policy layer encoding DGCA duty-of-care rules with machine-readable denial reasons.
adversarial dilation-shrink loss *Dr. Nirav Bhatt*	• Reimplemented **Dink-Net (ICML 2023**) graph **clustering** from scratch in **PyTorch: 2-layer GCN** encoder, self-supervised discrimination pretrain on corrupted view, adversarial **dilation-shrink loss** over K trainable centers • Diagnosed a gradient-conflict **bug degrading** cluster quality below the pretrained baseline; fixed it via dual-optimizer separation **(cluster centers 1e-2, encoder 1e-3)** and confirmed the root cause via targeted ablation. • Built a Hungarian-matched **ACC/NMI/ARI/F1** eval harness benchmarking **K-Means, Spectral**, and **GMM**; hit **97.75%** ACC on Wine with **mini-batch O memory** independent of graph size, scaling to **100M+** node graphs.
Voice Assistant Pipeline *(STT + LLM + TTS)*	• Built a **full-duplex voice** assistant streaming browser mic audio through local **Whisper STT**, **Gemini LLM**, and TTS to playback over a **single WebSocket**, with per-stage **latency instrumented** and **surfaced live** in the UI. • Cut time-to-first-audio **26% (8.95s to 6.63s, median 1.36s saved per turn**) by **chunking** the **LLM token** stream at sentence **boundaries and dispatching** each finished sentence to **TTS** while **later ones still generated** • Benchmarked all **three Whisper model variants** to document the accuracy versus latency tradeoff, measuring **base.en** at 10.5x faster than the **large-v3-turbo** default at identical word error rate, clean synthetic test audio.
P&I application	• Contributed to P&I App, whose website is used by **1,000+ users** through technical work and project coordination. • Boost **efficiency by 30%** through structured task delegation and clear communication of **6-member** WebOps team.
Fanatics Mock Trading	• Achieved **18% portfolio** growth by **executing data-driven** trades across 10+ equity and options instruments. • Used technical indicators like **RSI, MACD**, and **Moving Averages** to support trade decisions and risk analysis.
Flooding and flood mitigation	• Identified major flood-prone zones in IITM using historical rainfall and terrain data under Prof. Balaji Narasimhan. • Designed and simulated an **underground pipe network** for a high-risk area to **test 10-year flood resilience**.
Twisting deformation	• Analyzed torsional behavior **of solid, annular, and slit** tubes using **derived equations** and FEA in **ANSYS**. • Modeled components in CAD and identified high-stress zones under real-world torque inputs (e.g., 1000 Nm). • Assessed impact of **cross-sectional geometry** on **stress concentration** and **twist angles.**
Note Cloud (Aug'24-Sept'24)	• Developed a **full-stack cloud-based note-taking app** using React for frontend, Node.js for backend, and MongoDB. • Implemented **JWT authentication**, encrypted passwords with bcryptjs, and validation with express-validator. • Secured API endpoints using **CORS headers** and middleware to ensure clean access control and request integrity. • Created a responsive UI in React with full **CRUD operations**, real-time updates, and live data sync using Mongoose..
Code Compiler (July'24-Aug'24)	• Built a **multi-language online compiler** integrating **CodeX API**, supporting **Java, C++, Python, and JavaScript**. • Designed a **Next.js-based** frontend for writing, executing, and displaying **real-time code output** directly browser. • Added support for **language selection**, **starter code snippets**, and **live user input/output configuration**. • Managed API interactions, **asynchronous functions** and **implemented detailed error handling mechanisms**.
News Monkey (June'24-July '24)	• Created a **news aggregation app** using Next.js with **server-side rendering** for SEO optimization and fast loading. • Fetched real-time headlines by integrating the News API, categorized by topic using async data fetching methods. • Built an SEO-friendly, mobile-optimized UI using **Tailwind CSS**, including **lazy loading** for media-heavy content. • Added search, dynamic routing, and detailed article pages to improve user accessibility and content engagement.
Stock Market Screener and Back tester	• Developed a stock screener and backtester using **Python**, leveraging libraries for real-time historical data handling. • Visualized strategy outcomes using **Matplotlib** for time series plots, enabling comparisons and signal validation. • Automated CSV export of stock data into organized folders using **os**, improving traceability and analysis efficiency. • Backtested **3 technical** strategies (Crossover, RSI, MA) across **5** Nifty 50 stocks, analyzing ~180 signal combinations. • Implemented SMA Crossover for trend detection; achieved up to **18%** returns in bullish markets vs. 9–12% baseline. • Evaluated strategies using metrics like **CAGR, Win Rate, and Sharpe Ratio**; organized results by stock and strategy
Placement Coordinator UIR (May'24-May'25)	• **Managed** outreach to 15+ Institutes, driving a increase in research Intern and improving intern offer conversion. • **Headed** a pipeline to partially automate institute lead generation via web scraping, streamlining outreach. • Compiled a structured Blue Book documenting 50+ research internship and placement records across departments, Improving data accessibility for students navigating the recruitment process.
Mini **Tic Tac Toe**	• Created a **Tic Tac Toe game** with custom logic for win/draw detection, responsive layout, and user interactions.
Mini **Stone Paper Scissors**	• Built an interactive **Stone Paper Scissors game** using **JavaScript** with gameplay and **DOM manipulation**. • Focused on **vanilla JS, responsive design principles**, and core web dev techniques without external frameworks.
Mini **Odd Even**	• Designed a lightweight **Odd/Even checker** tool with **real-time input response** and intuitive, **minimalist UI**.

RELEVANT COURSEWORK AND SKILLS

Introduction to Programming	Probability and Statistics	Linear Algebra
Mechanics of Materials	Engineering Design	Programming and DSA in Python*
Series and Matrices	Functions of Several Variables	Financial and Equity Market Analyst

EXTRA-CURRICULAR ACTIVITIES

Social Work	**Volunteer** Placement Team	• **Coordinated** and facilitated internship and placement tests, ensuring efficient scheduling and smooth execution • Maintained strict exam integrity while acting as an **invigilator**, consistently upholding academic standards
	Samruddhi Finance Club	• Conducted on-ground workshops for **11th** and **12th** grade students in **Pondicherry** and areas around Chennai. • Targeted economically **disadvantaged students**, educating them on identifying and avoiding financial fraud.
	SMC[3]	• Counselled **4** first-year students as a **Saathi mentor** on issues about academic & extra-curricular activities
	Workshop	• Educated **50+ participants** on **finance fundamentals** by conducting an interactive workshop during Shaastra.
NCC[5]	Cadet Corporal B-certificate Holder[4]	• Developed **leadership skills** by leading teams in various drills and exercises during NCC activities. • Enhanced **teamwork** and **communication** skills by collaborating with peers in multi-disciplinary events and camps. • Completed specialized training in first aid, emergency response, and disaster management as part of NCC curriculum.
Abacus | Level 2		• Qualified **Level 2** of Abacus with top performer while solving **100 questions per minute**
Sports | Yellow Belt		• Gain a Yellow belt in karate and **3 gold medals** at the school level, and **2 times runner-up** at the district level

1. Centre for Innovation Hub | 2. Financial Fest of IIT Madras | **3.** Saathi Mentorship Cell | 4. To declare the results delayed due to tensions | 5. National Cadet Corp